Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

INSTALLER GUIDE · BATTERY COMMUNICATION

Deye: parallel battery communication

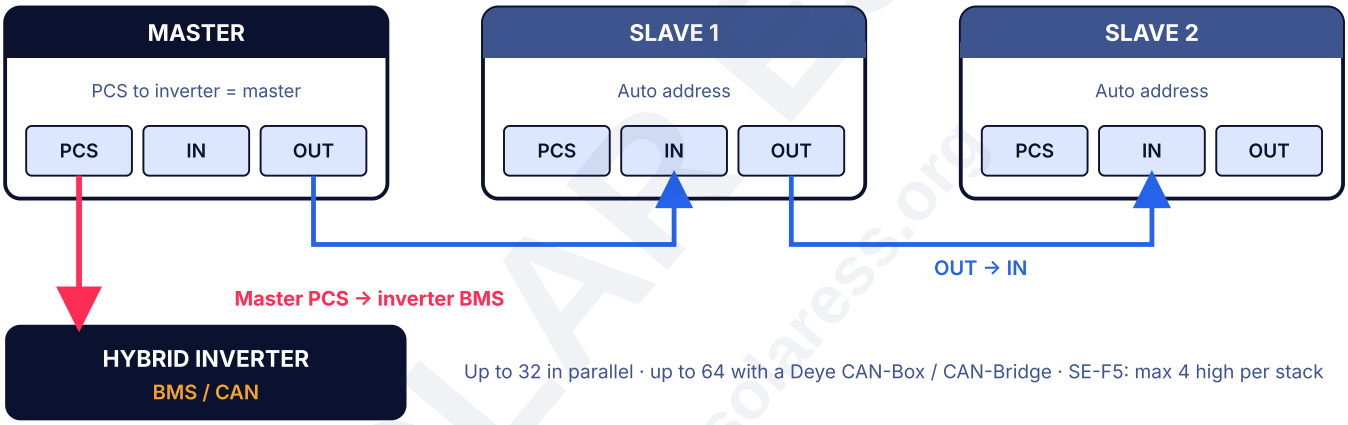
SE-G5.1 Pro · SE-F5 · SE-F12 · RW-M6.1 · RW-F10.2 low-voltage batteries (51.2 V)

One rule for every brand: only the **master** battery talks to the inverter. All other batteries are chained to it.

1 · Set the master / address

No DIP switch. Current Deye LV batteries auto-network their addresses. The battery whose **PCS** port is cabled to the inverter becomes the master; the others become slaves through the IN / OUT chain. (Some very early SE-G5.1 Pro units had an address dial: if yours has one, follow its manual.)

2 · Connect the communication cables



1. Inverter **BMS** port → **Master PCS** port (CAN 500 kbps).
2. **Master OUT** → **Slave 1 IN**, then Slave 1 OUT → Slave 2 IN, and so on. Leave the last OUT empty.
3. Never cross PCS and IN / OUT cables: they have different pinouts. PCS: CAN-H pin 4, CAN-L pin 5 (matches the Deye BMS port, so a straight cable works).
4. Power: equal-length, equal-size cables to a busbar. If only the first battery feeds the inverter, respect its current limit (SE-F5 / SE-G5.1 Pro: 120 A; F12 / RW-F10.2: 230–250 A).

SOLAR ESS www.solaress.org

Solar ESS Co. Ltd.
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
Tel 063 151 6694 · 089 632 5829
www.solaress.org · www.โซลาร์เอสส์.com

3 · Inverter settings

- Battery Setting → Batt Mode = **Lithium**.
- Lithium Mode = **00** (CAN) per the Deye approved battery list.
- Tick **BMS_Err_Stop** so the inverter stops if BMS communication is lost (F58).
- Check SOC, voltage and charge/discharge current limits on the inverter screen. Re-check charge settings after any firmware update.

4 · Power-on sequence

1. Check all cables. Close the battery DC breaker.
2. Switch on the BMS / power button on each battery (SE-G5.1 Pro: green RUN flashing = normal).
3. Start the inverter. (RW-F10.2-B manual: inverter first, then battery breaker, then battery button.)
4. Confirm the inverter reads SOC and the correct total capacity.
5. Shut down in reverse order.

Match battery voltage / SOC before paralleling. With non-Deye inverters, limit the current (SE-F5: 100 A charge / 120 A discharge).

5 · Cable & good practice

Ports	PCS = inverter only. IN / OUT = battery-to-battery only. USB = firmware updates.
Cable	Standard straight LAN cable (CAT5e/CAT6, T568B both ends). No crossover.
Routing	Keep communication cables away from DC power cables. Plugs must click in fully.
Mixing	Never mix brands, models or capacities in one battery bank or one communication chain.
Voltage	Before paralleling, charge or rest all packs so their voltages are close.

6 · Troubleshooting: "no SOC" or "BMS communication fail"

F31 · PCS communication fault	Check the master PCS → inverter BMS cable and Lithium Mode 00.
F30 · internal communication fault	Check the OUT → IN cables between batteries.
F32 · host address duplication	Two batteries think they are master: the PCS or OUT wiring is wrong.
Inverter F58	BMS communication lost: check the PCS cable and battery power.
Buzzer every 2 seconds	SOC at or below 5%: charge the battery.

SOLAR ESS www.solaress.org